EV Technology Course — Module 14: Electric Two- and Three-Wheelers

Oct 14, 2026 · @Ashishpreet

In India, electrification is led by two- and three-wheelers, not cars. India sold about 1.4 million electric two-wheelers in FY2026, roughly 6.5% of all two-wheelers, and electric three-wheelers already form a large share of new three-wheeler registrations. These vehicles use the same principles as electric cars but with very different cost, packaging and duty-cycle constraints.

1. Learning objectives

- Describe the architecture of an electric scooter, motorcycle, e-rickshaw and e-auto.
- Compare hub motors and mid-drive motors.
- Explain battery choices (NMC, LFP, lead-acid) and battery swapping.
- Calculate range, motor torque and running cost for small EVs.
- Understand Indian categories (L1, L2, L3, L5), regulations and incentives.
- Identify cost and quality issues specific to small EVs.

2. Fundamentals

- **Indian categories:** L1 (low-speed, ⩽25 km/h and ⩽250 W, no registration for some), L2 (two-wheelers above that), L3 (e-rickshaw/e-cart, ⩽25 km/h), L5 (three-wheelers like e-autos, higher speed).
- **Typical e-scooter:** 2–4 kWh battery, 3–7 kW peak motor, 48–72 V system, 25–35 Wh/km consumption, 80–150 km real range.
- **Typical e-auto (L5):** 7–10 kWh LFP, 48–60 V or higher, 9–11 kW motor, 60–80 Wh/km.
- **E-rickshaw (L3):** 48–60 V, lead-acid or LFP, 1–2 kW motor, very low cost.
- Most small EVs use **low voltage (<60 V)**, so HV isolation rules are simpler than in cars, but currents are high.

3. Detailed technical explanation

Hub motor vs mid-drive

	Hub motor	Mid-drive (frame-mounted)
Location	Inside the rear wheel	On the frame, drives wheel by belt, chain or gear
Gearing	Direct drive, no reduction	Reduction (e.g. 6–10:1)
Torque at motor	High (big, heavy motor)	Low (small, fast motor)
Unsprung mass	High (worse ride and handling)	Low
Cost and simplicity	Cheaper, fewer parts	More parts but better performance
Examples	TVS iQube, many entry e-scooters, e-rickshaws	Ather, Ola S1, Ultraviolette F77 (motorcycle)

Battery choices

- **NMC cylindrical cells (18650/21700)** in modules: high energy density, compact, but need careful thermal design; common in premium scooters.
- **LFP prismatic or cylindrical:** safer, longer life, heavier; common in three-wheelers and increasingly in scooters.
- **Lead-acid:** cheapest, heavy, short life; still used in e-rickshaws, being replaced by LFP.
- **Portable batteries:** removable packs for charging at home (e.g. the Honda Activa e: uses swappable packs, while most Indian scooters have a fixed pack).

Battery swapping

- Drained battery is exchanged for a charged one at a station in 1–2 minutes; users pay per swap or subscription.
- Operators: Battery Smart, Sun Mobility, Gogoro (with partners), Honda's swapping network for the Activa e: in Bengaluru.
- Benefits: low upfront cost (battery as a service), no waiting; challenges: standardisation, capital for battery inventory, BMS data and ownership rules.

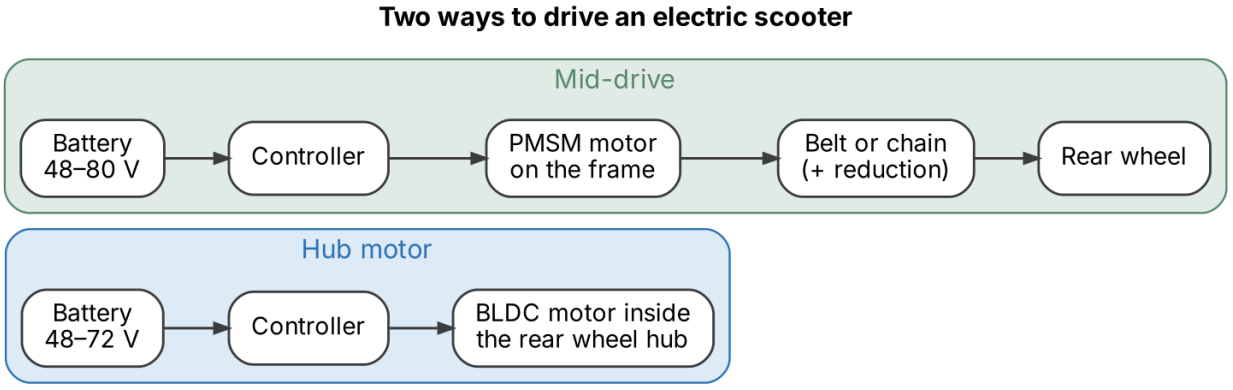
Controller and electronics

- Motor controller (MOSFET-based inverter) for BLDC/PMSM, BMS, vehicle control unit, dashboard with connectivity (TCU, app).
- Regenerative braking is limited by battery size and rider comfort; combined braking systems (CBS) are mandatory on many small two-wheelers.

Regulation and incentives (India)

- Battery safety: **AIS-156** (L category), introduced and tightened after 2022 scooter fires.
- **PM E-DRIVE:** e2W subsidy of ₹2,500 per kWh, capped at ₹5,000 per vehicle (and 15% of ex-factory price), for registrations from April 2025, extended to March 2028 per a notification reported in August 2026. The L5 e3W subsidy closed after the segment hit its target in December 2025.
- **PLI Auto** rewards local value addition; domestic value addition must be verified by testing agencies.

4. Diagram: hub motor vs mid-drive scooter



5. Step-by-step: how an e-scooter accelerates

1. The rider twists the throttle; a hall-effect sensor sends a signal to the controller.
2. The controller (VCU logic inside or separate) chooses the torque based on ride mode (Eco, Ride, Sport) and battery state.
3. The BMS confirms allowable discharge current.
4. The controller switches MOSFETs to drive current into the motor phases.
5. The motor produces torque at the wheel directly (hub) or through a belt/gear (mid-drive).
6. On release or braking, the controller commands regen, limited by battery charge acceptance.
7. Dashboard and app show range, SOC and ride data.

6. Engineering equations

Range:

$$\text{Range} = \frac{E_{usable}}{\text{Wh/km}}$$

Wheel torque requirement:

$$T_{wheel} = (m\,a + m\,g\,C_{rr}) \cdot r_{wheel}$$

Motor torque with reduction ratio G:

$$T_{motor} = \frac{T_{wheel}}{G \cdot \eta}$$

Running cost:

$$\text{₹/km} = \text{Wh/km} \times \frac{\text{₹/kWh}}{1000}$$

7. Worked examples

Example 1: scooter range. A 3 kWh scooter with 90% usable energy (2.7 kWh) at 30 Wh/km: **90 km**.

Example 2: running cost. 30 Wh/km × ₹8/kWh = **₹0.24/km**. A petrol scooter at 50 km/L with petrol at ₹95/L costs **₹1.90/km**. For 40 km per day, annual saving ≈ (1.90 − 0.24) × 40 × 365 ≈ **₹24,000**.

Example 3: hub vs mid-drive torque. 150 kg (scooter + rider), accelerating at 1.5 m/s² with Crr 0.01: force ≈ 240 N. With a 0.2 m wheel radius, wheel torque ≈ **48 Nm**. A hub motor must produce all 48 Nm; a mid-drive with 10:1 reduction needs only about **4.8 Nm** (before losses), allowing a much smaller motor.

Example 4: e-rickshaw daily energy. 120 km/day at 60 Wh/km = **7.2 kWh/day**.

8. Real-world examples

- **FY2026 e2W leaders (Vahan-based):** TVS (~24% share), Bajaj (~21%), Ather (~17%), Ola Electric (~12%), Hero Vida (~10%); total ~1.40 million units, up 22% year on year.
- **Ola Electric** built a large Krishnagiri "Futurefactory" and is building cell capacity; its share fell sharply in FY2026 amid service and quality complaints.
- **Ather** uses in-house-designed packs and a mid-mounted PMSM with belt drive; its Ather Grid charging network is open to other brands.
- **Mahindra Last Mile Mobility, Bajaj, Piaggio, Kinetic Green** and many smaller makers lead e3W; e-rickshaws (L3) are a huge, largely unorganised market.
- **2025 rare-earth export curbs** from China disrupted magnet supply for many Indian two-wheeler makers, pushing interest in ferrite and rare-earth-free motors.

9. Manufacturing applications

- **Cell sourcing and pack assembly:** cylindrical cells welded with nickel strips or laser-welded busbars; potting and IP67 sealing.
- **Motor manufacturing:** outer-rotor hub motors (wound stator, magnet ring) or compact mid-drive PMSMs.
- **Frame:** steel tube or aluminium die-cast frames; plastic body panels (injection moulding: PP, ABS, PC-ABS).
- **EOL testing:** BMS communication, insulation, roller test, brake test, software flashing.
- **Cost:** battery 35–45% of an e-scooter's cost; motor + controller 12–18%; body plastics and frame 15–20%.
- **Quality:** water ingress, connector failure, thermal events and software bugs are the main field issues.

10. Common mistakes and misconceptions

- **"Low voltage means no safety issue."** High currents and dense cells can still cause fires.
- **"Claimed IDC range is real."** Real range is often 20–40% lower.
- **"Swapping solves everything."** It needs standard batteries, dense networks and capital.
- **"Hub motors are always worse."** They are cheap and simple, good for low-cost scooters and e-rickshaws.

11. Practical case study: scooter battery fire

Problem. Several scooters of one brand catch fire in summer 2022.

Root causes found by the expert committee: cell defects, weak BMS protection, poor module design allowing cell-to-cell propagation, and inadequate venting.

Response. Recalls; government amended AIS-156 and AIS-038 with thermal propagation tests, cell-level testing, BMS requirements and IP ratings; OEMs improved cell sourcing and pack design.

Lesson. Cost-down cannot compromise cell quality, BMS design and validation.

12. Revision notes

- Categories: L1, L2, L3 (e-rickshaw), L5 (e-auto).
- Hub motor = cheap and simple; mid-drive = lighter wheel, smaller motor via gearing.

- NMC for compact premium packs; LFP for safety and life; lead-acid fading.
- Swapping: Battery Smart, Sun Mobility, Gogoro, Honda.
- India FY2026: ~1.4 million e2Ws, ~6.5% penetration.

13. Glossary

Term	Meaning
L3 / L5	Indian three-wheeler categories (low-speed / higher-speed)
Hub motor	Motor built into the wheel
Mid-drive	Frame-mounted motor driving wheel through reduction
BaaS	Battery as a service
CBS	Combined braking system
IDC	Indian driving cycle used for range claims
Vahan	Government vehicle registration database

14. Knowledge check

Conceptual

1. Why are two- and three-wheelers leading EV adoption in India?
2. Two drawbacks of hub motors?
3. Why is LFP popular in e-autos?
4. What problem does battery swapping solve?
5. Why were battery standards tightened in 2022?

Numerical

1. 2.5 kWh usable, 28 Wh/km. Range?
2. An e-auto has 8 kWh usable LFP at 70 Wh/km. Range?
3. If a scooter uses 30 Wh/km and electricity costs ₹10/kWh, cost per km?

Application

1. As a costing engineer, list the top five cost items in an e-scooter BOM and one VA/VE idea for each.

2. Should an e-auto fleet operator choose fixed battery or swapping? Give the main factors.

15. Answer key

Conceptual

1. Huge existing market, short daily trips, low running cost advantage, lower battery size needed, incentives.
2. High unsprung mass (worse ride) and large, heavy, costly motor for the torque; also limited cooling and exposure to road shocks and water.
3. Safety, long cycle life for high daily kilometres, lower cost per kWh.
4. Long charging time and high upfront battery cost.
5. A string of scooter fires showed weak cell, BMS and pack design.

Numerical

1. 2,500 ÷ 28 = **89 km**.
2. 8,000 ÷ 70 = **114 km**.
3. 30 × 10 ÷ 1,000 = **₹0.30/km**.

Application

1. Battery pack (LFP or larger-format cells, cell-to-pack), motor (ferrite or reduced-magnet design), controller (integrated motor-controller), plastics (part consolidation, material grade rationalisation, family moulds), frame (tube optimisation or die-cast simplification).
2. Swapping suits high daily kilometres and dense swap networks with no downtime; fixed battery suits predictable routes and depot charging with lower per-km cost. Consider network coverage, swap price per km, battery ownership, resale and downtime.

16. Further learning

- Vahan dashboard (vahan.parivahan.gov.in) for monthly registrations.
- Ministry of Heavy Industries PM E-DRIVE guidelines.
- Autocar Professional and EVreporter for e2W/e3W analysis.
- NITI Aayog battery swapping policy draft (2022).

Sources

- Record 1.4 million electric 2Ws sold in FY2026 — Autocar Professional
- Govt extends PM E-Drive subsidy for e2Ws till FY28 — Autocar Professional, 11 Aug 2026

Next: Module 15, commercial EVs, buses, trucks and fuel cells.